

A Topology-Driven Invariant Mass Kernel in Swirl-String Theory

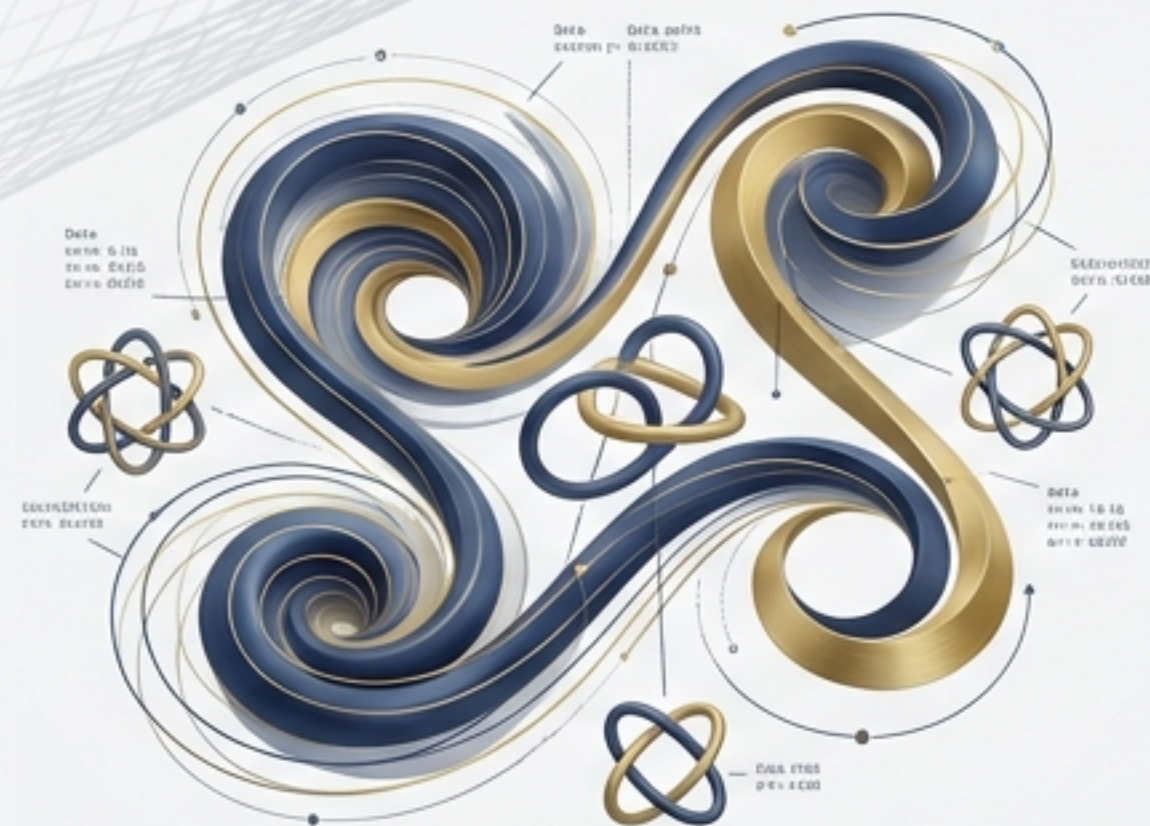
Reference Implementation and Benchmark Against Atomic and Molecular Masses

Model: Swirl-String Theory (SST)

Focus: Invariant Mass Kernel

Scope: Methodology + Benchmark Results

Source Code: SST_Atom_Mass_Invariant.py



The Central Hypothesis: Mass Emerges from Topology



T

Braid Index (b_T)
Seifert Genus (g_T)
Component Count (n_T)
Total Ropelength ($L_{\text{tot}}(T)$)

SST Invariant
Mass Kernel



$M(T)$

The Swirl–String Theory (SST) kernel provides a fixed, mode-independent functional that maps a set of dimensionless topological invariants of an object T to its inertial mass $M(T)$.

Deconstructing the Invariant Mass Kernel

****Braid Penalty***

Exponential term scaling with braid complexity (b_T) and the fine-structure constant (α).

****Genus Penalty****

Damping factor based on topological genus (g_T) and the golden ratio (ϕ).

$$M(T) = \left(\frac{4}{\alpha}\right)^{bT-3/2} * \phi^{-gT} * nT^{-1/\phi} * u \frac{V(T)}{c^2}$$

****Component Penalty****

Damping factor based on the number of string components (n_T).

****Core Mass Contribution****

The fundamental mass derived from energy density (u) and effective volume ($V(T)$).

An Intuitive Model for the Core Mass Contribution

$$\text{Mass} \propto (\text{Energy Density}) \times (\text{Effective Volume})$$



$$u = \frac{1}{2} * \rho_{core} * v_{\odot}^2$$

****Characteristic Kinetic Energy Density****

Think of this as how 'packed' the motion-energy is.



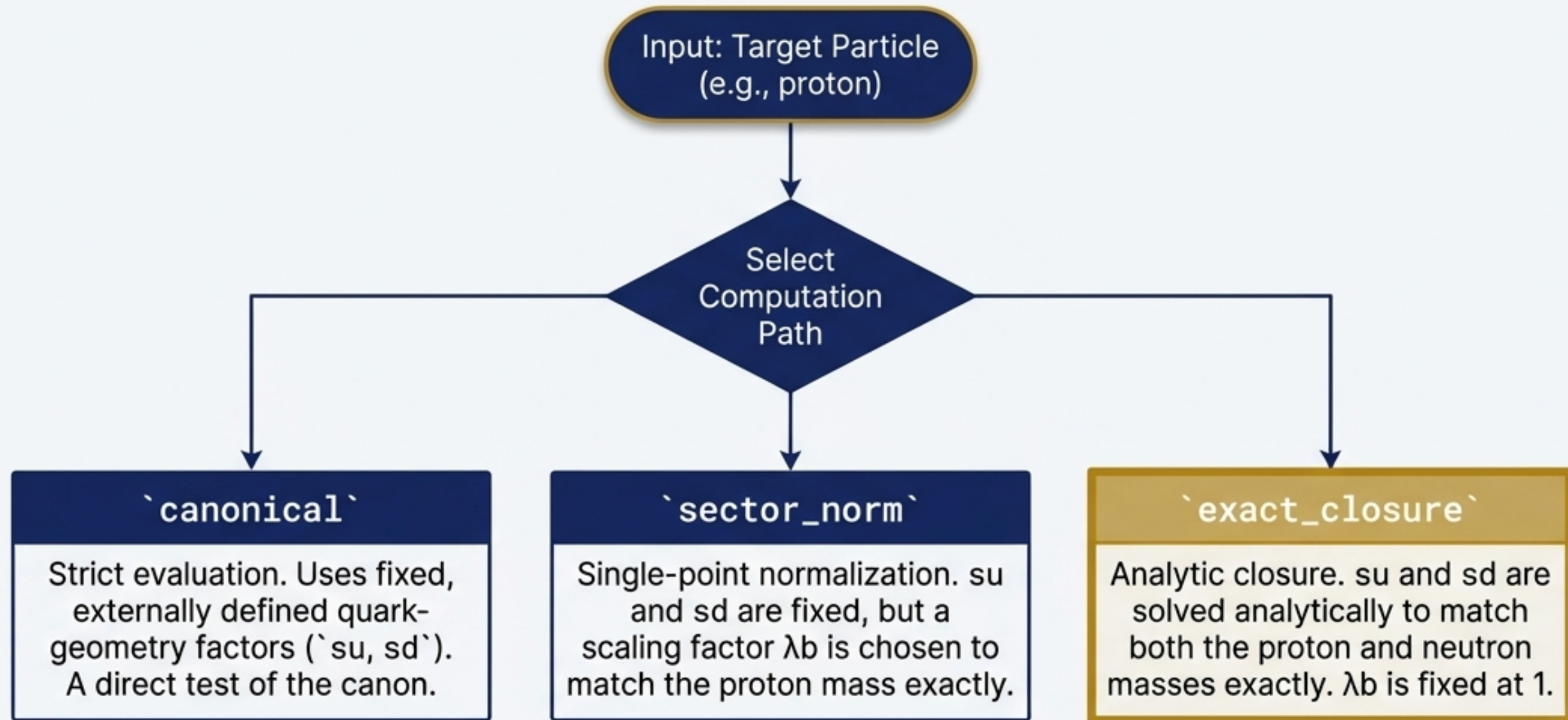
$$V(T) = \pi * r_c^3 * L_{tot}(T)$$

****Effective Volume Proxy****

And this is how much of that packed region the string-shape occupies.

“more packed energy times more occupied volume means more mass.”

Three Computational Paths for Applying the Kernel



The benchmark results presented in this deck are generated using the `exact_closure` mode.

Model Calibration: Setting the Geometric Inputs



Lepton Calibration



For a given target mass (M_{target}) and a fixed base topology (b, g, n), the model solves for the required dimensionless ropelength $L_{\text{tot}}(T)$.

Base Topologies

Electron: $(b, g, n) = (2, 1, 1)$

Muon: $(b, g, n) = (5, 2, 1)$

Tau: $(b, g, n) = (7, 3, 1)$

This step makes the listed lepton masses exact by construction; it is a calibration, not a prediction.



Baryon 'exact_closure'



For a fixed baryon topology of $(3, 2, 3)$, the model solves a system of linear equations to find the dimensionless geometric factors su and sd that exactly reproduce the measured proton and neutron masses.

$$M_p = K(2su + sd)$$

$$M_n = K(su + 2sd)$$

This ensures the model's building blocks (proton, neutron, electron) are perfectly matched before assembling larger atoms.

Derived Geometric Factors and Calibrated Ropelengths

Numerical results from the `exact\_closure` implementation

1. Baryon Geometric Factors

$$su = 3.03060$$

$$sd = 3.04315$$

(Both values are dimensionless)

2. Calibrated Ropelengths `Ltot(T)`

$$L_{\text{tot}}(\text{electron}) = 3.33963 \times 10^{-2}$$

$$L_{\text{tot}}(\text{muon}) = 4.41651 \times 10^1$$

$$L_{\text{tot}}(\text{tau}) = 1.99071 \times 10^3$$

$$L_{\text{tot}}(\text{proton}) = 3.59425 \times 10^2$$

$$L_{\text{tot}}(\text{neutron}) = 3.59921 \times 10^2$$

3. Key Ratios

$$L(\text{muon}) / L(\text{electron}) \approx 1.322 \times 10^3$$

$$L(\text{tau}) / L(\text{muon}) \approx 4.507 \times 10^1$$

Assembling Atoms: From Nucleons to Nuclides

Step 1: Sum of Constituents



$$M_{\text{sum}}(Z, N) = Z \cdot M_p + N \cdot M_n + Z \cdot M_e$$

Begin with the sum of the predicted masses of the constituent protons, neutrons, and electrons for a nuclide (Z, N).

Step 2: Binding Energy Correction



$$- \Delta m(Z, N)$$

Subtract a mass defect Δm calculated from a semi-empirical model for nuclear binding energy.

Step 3: Predicted Atomic Mass



$$M_{\text{pred_atom}}(Z, N) = M_{\text{sum}} - \Delta m$$

The result is the final predicted mass for the atom.

The Verdict: Benchmark Summary Statistics

Model performance in `exact\_closure` mode across 92 elements (H-U)

Mean Absolute Error

$1.960 \times 10^{-1} \%$

Mean Absolute Percentage Error | $\text{MAE}(|\epsilon|)$

Median Absolute Error

$6.900 \times 10^{-2} \%$

Median Absolute Percentage Error | $\text{median}(|\epsilon|)$

Maximum Error

1.840%

Maximum Absolute Percentage Error | $\text{max}(|\epsilon|)$

RMSE

$3.740 \times 10^{-1} \%$

Root Mean Square Error | $\text{RMSE}(|\epsilon|)$

Percentage error ' ϵ ' is defined as $100\% \times (M_{\text{pred}} - M_{\text{ref}}) / M_{\text{ref}}$.

Model Performance on Selected Atoms and Molecules

Object	Actual Mass (kg)	Predicted Mass (kg)	Percentage Error ($\epsilon\%$)	Error Visual
Hydrogen (H)	1.67356×10^{-27}	1.67353×10^{-27}	-0.00148	
Carbon (C)	1.99447×10^{-26}	1.99310×10^{-26}	-0.0688	
Iron (Fe)	9.27328×10^{-26}	9.28766×10^{-26}	+0.155	
Gold (Au)	3.27076×10^{-25}	3.27062×10^{-25}	-0.00435	
Uranium (U)	3.95258×10^{-25}	3.95269×10^{-25}	+0.00285	
Water (H ₂ O)	2.99146×10^{-26}	2.99102×10^{-26}	-0.0148	
Carbon Dioxide (CO ₂)	7.30803×10^{-26}	7.30573×10^{-26}	-0.0316	
Caffeine (C ₈ H ₁₀ N ₄ O ₂)	3.22460×10^{-25}	3.22341×10^{-25}	-0.0369	

Actual mass (M_{ref}) is derived from standard molar mass (M_{mol}) via the Avogadro constant (N_A).

A Closer Look at the Largest Deviations

Highlighting the five elements with the largest absolute percentage error in the benchmark set.

Object	Actual Mass (kg)	Predicted Mass (kg)	Percentage Error ($\epsilon\%$)
Boron (B)	1.79504×10^{-26}	1.82806×10^{-26}	+1.839
Chlorine (Cl)	5.88661×10^{-26}	5.80622×10^{-26}	-1.366
Magnesium (Mg)	4.03594×10^{-26}	3.98256×10^{-26}	-1.323
Lithium (Li)	1.15241×10^{-26}	1.16499×10^{-26}	+1.091
Neon (Ne)	3.35097×10^{-26}	3.31944×10^{-26}	-0.941

The largest deviations provide critical data points for refining the model, particularly the binding energy corrections.

Understanding the Model's Scope and Limitations



Isotope Mismatch

The benchmark compares predictions for a single nuclide (Z, N) against standard atomic weights, which are often isotopic averages. A strict test requires isotope-specific mass data (e.g., AME tables).



Semi-Empirical Binding Model

The nuclear binding energy correction (Δm) is based on the phenomenological Semi-Empirical Mass Formula (SEMF), not derived from SST. The accuracy reflects the combination of the kernel and this empirical proxy.



Lepton Ropelengths are Calibrated

Muon and tau ropelengths are solved to fit their known masses. They are not predictions unless an independent SST rule for lepton L_{tot} is introduced.

A Path Forward: Two Immediate Falsifiable Checks



Test the Canonical Baryons

Hypothesis

The core SST canon can predict nucleon masses without fitting.

Method

Execute the `'canonical'` path. Fix the geometric factors (su , sd) from external hyperbolic-volume assignments and set $\lambda_b = 1$.



Outcome

The resulting residuals for the proton and neutron masses become a direct, uncalibrated test of the model's foundational mapping.

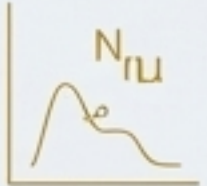
Isotope-Resolved Benchmark

Hypothesis

Model accuracy will improve when compared to precise isotopic masses.

Method

Replace the standard atomic weight reference table with isotope-specific masses for the chosen (Z, N) nuclides.



Outcome

Re-evaluating the error statistics against this 'cleaner' dataset will isolate the kernel's performance from isotopic averaging effects.

This implementation provides a transparent and reproducible framework for testing these and other core tenets of Swirl-String Theory.